

Q.2

Solⁿ:-

Let $u = c_1 u_1 + c_2 u_2 + \dots + c_n u_n$, $c_i \in \mathbb{R}$

To show that $u = 0$, we show that $\|u\|^2 = 0$

$$\begin{aligned} \Rightarrow \langle u, u \rangle &= \langle c_1 u_1 + \dots + c_n u_n, c_1 u_1 + \dots + c_n u_n \rangle \\ &= c_1 \langle u_1, u_1 \rangle + \dots + c_n \langle u_n, u_n \rangle \\ &= 0 \end{aligned} \quad [\because \langle u, u_i \rangle = 0 \text{ for } i=1, \dots, n]$$

Q.3

Solⁿ:-

Set $x_1 u_1 + x_2 u_2 + x_3 u_3 = u$

Since $u_i \perp u_j$, $i \neq j$

$$\begin{aligned} \Rightarrow \langle u, u_i \rangle &= \langle x_1 u_1 + x_2 u_2 + x_3 u_3, u_i \rangle \\ &= x_i \langle u_i, u_i \rangle, \quad i=1, 2, 3. \end{aligned}$$

$$\Rightarrow x_i = \frac{\langle u, u_i \rangle}{\langle u_i, u_i \rangle}$$

$$\Rightarrow x_1 = \frac{7+2+9}{1+4+1} = 3, \quad x_2 = -1, \quad x_3 = 2.$$

Solⁿ 1 $\Rightarrow$ (i) Linearity:

$$\begin{aligned} \langle f+g, h \rangle &= \int_a^b (f+g) h(x) dx \\ &= \int_a^b f h dx + \int_a^b g h dx \\ &= \langle f, h \rangle + \langle g, h \rangle \end{aligned}$$

(ii) Homogeneity:

$$\begin{aligned} \langle \alpha f, g \rangle &= \int_a^b (\alpha f) g dx \\ &= \alpha \int_a^b f g dx = \alpha \langle f, g \rangle \end{aligned}$$

(iii) Symmetry: $\langle f, g \rangle = \langle g, f \rangle$

Similarly, we can show that:

$$\langle f, f \rangle = \int_a^b f^2(x) dx \geq 0 \quad (\because f^2 \geq 0)$$

$$\& \quad \langle f, f \rangle = 0$$

$$\Rightarrow f^2 = 0$$

$$\Rightarrow \boxed{f = 0}.$$

→ Gram-Schmidt process → Let V be an I.P.S and $S = \{x_1, x_2, \dots, x_n\}$ be a L.I. set of V . Then we ~~obtain~~ obtain an orthogonal set $S' = \{y_1, y_2, \dots, y_n\}$ s.t. $\text{span}(S) = \text{span}(S')$

where $y_1 = x_1$ and $y_k = x_k - \sum_{i=1}^{k-1} \frac{\langle x_k, y_i \rangle}{\|y_i\|^2} y_i$ for $2 \leq k \leq n$.

or $y_k = x_k - \sum_{i=1}^{k-1} \text{proj}_{y_i}(x_k)$, $2 \leq k \leq n$.

And if take $z_k = \frac{y_k}{\|y_k\|}$, then $\{z_1, z_2, \dots, z_n\}$ is orthonormal set.

Ques-7 (i) Given that

$$B = \{ (1, 2, -2), (2, 0, 1), (1, 1, 0) \}$$

is a basis of $\mathbb{R}^3(\mathbb{R})$.

We need to convert B as an orthogonal basis by using Gram-Schmidt process.

$$\text{Let } x_1 = (1, 2, -2)$$

$$x_2 = (2, 0, 1),$$

$$x_3 = (1, 1, 0)$$

and suppose $B' = \{ y_1, y_2, y_3 \}$ is the required orthogonal basis.

then by Gram-Schmidt process

~~then~~ $y_1 = x_1 = (1, 2, -2)$

$$y_2 = x_2 - \frac{\langle x_2, y_1 \rangle}{\|y_1\|^2} y_1$$

$$= (2, 0, 1) - (1, 2, -2) = (1, -2, 3)$$

$$= (2, 0, 1) - \frac{2-2}{9} (1, 2, -2) = (2, 0, 1)$$

$$\text{and } y_3 = x_3 - \frac{\langle x_3, y_1 \rangle}{\|y_1\|^2} y_1 - \frac{\langle x_3, y_2 \rangle}{\|y_2\|^2} y_2$$

$$= (1, 1, 0) - \frac{1+2}{9} (1, 2, -2) - \frac{2}{5} (2, 0, 1)$$

$$= (1, 1, 0) - \left(\frac{1}{3}, \frac{2}{3}, -\frac{2}{3}\right) - \left(\frac{4}{5}, 0, \frac{2}{5}\right) = \left(-\frac{2}{15}, \frac{5}{15}, \frac{4}{15}\right)$$

Hence required orthogonal basis is $\{ (1, 2, -2), (2, 0, 1), \left(-\frac{2}{15}, \frac{5}{15}, \frac{4}{15}\right) \}$

(ii) Same as (i).

Ques - 8 → We need to find an orthonormal basis which contain (i) the vector $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0)$

Suppose $B = \{(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0), (1, 0, 0), (0, 0, 1)\}$

Then say $\alpha_1 = (\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0)$, $\alpha_2 = (1, 0, 0)$

If $B' = \{y_1, y_2, y_3\}$ is the orthonormal basis then by Gram-Schmidt process.

Ques - 8 → We need to find an orthonormal basis of $\mathbb{R}^3$ which contain (i) the vector $(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0)$.

Suppose $B = \{(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0), (1, 0, 0), (0, 0, 1)\}$ be any basis.

Then say $\alpha_1 = (\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0)$, $\alpha_2 = (1, 0, 0)$, $\alpha_3 = (0, 0, 1)$

If $B' = \{y_1, y_2, y_3\}$ is the orthogonal basis

then by Gram-Schmidt process.

$$y_1 = \alpha_1$$

$$y_2 = \alpha_2 - \frac{\langle \alpha_2, y_1 \rangle}{\|y_1\|^2} y_1$$

$$= (1, 0, 0) - \frac{\frac{1}{\sqrt{2}}}{1} (\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0) = (\frac{\sqrt{2}-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0)$$

$$\text{and } y_3 = \alpha_3 - \frac{\langle \alpha_3, y_1 \rangle}{\|y_1\|^2} y_1 - \frac{\langle \alpha_3, y_2 \rangle}{\|y_2\|^2} y_2$$

$$= (0, 0, 1) - 0 - 0 = (0, 0, 1)$$

So $B' = \{(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0), (\frac{\sqrt{2}-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0), (0, 0, 1)\}$ is the orthogonal basis of $\mathbb{R}^3(\mathbb{R})$.

$$\text{So } B' = \{(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0), (\frac{1}{2}, \frac{3}{2}, 0), (0, 0, 1)\}$$

So $B' = \left\{ \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0 \right), \left(\frac{1}{2}, \frac{3}{2}, 0 \right), (0, 0, 1) \right\}$

is the orthogonal basis.

Now for orthonormal.

$$z_1 = \frac{y_1}{\|y_1\|} = \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0 \right)$$

$$z_2 = \frac{y_2}{\|y_2\|} = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right)$$

$$z_3 = \frac{y_3}{\|y_3\|} = (0, 0, 1)$$

Hence the required orthonormal basis is $\left\{ \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0 \right), \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right), (0, 0, 1) \right\}$

(ii) Same as (i),

just take $B = \left\{ \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}} \right), \left(\frac{2}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right), (0, 1, 0) \right\}$.

Ques - (16) (i) Given that

$$S = \{ (1, 1, 0, 1), (1, -2, 0, 0), (1, 0, -1, 2) \}$$

It can easily verify that S is L.I.

if ~~x_1, x_2~~ , $x_1 = (1, 1, 0, 1)$, $x_2 = (1, -2, 0, 0)$

$x_3 = (1, 0, -1, 2)$

then we can find $\rightarrow$ orthogonal vectors y_1, y_2, y_3 by Gram-Schmidt process as previously.

And then ~~z_1, z_2, z_3~~ we can find orthonormal vectors z_1, z_2, z_3 as $z_1 = \frac{y_1}{\|y_1\|}$, $z_2 = \frac{y_2}{\|y_2\|}$, $z_3 = \frac{y_3}{\|y_3\|}$

And we know that

if $S = \{ x_1, x_2, x_3 \}$

$S' = \{ y_1, y_2, y_3 \}$ and $S'' = \{ z_1, z_2, z_3 \}$

Then $\text{span } S = \text{span } S' = \text{span } S''$

Hence S'' will be the orthonormal basis of the subspace of $\mathbb{R}^4(\mathbb{R})$ which is spanned by S .